

5G RAN

Beam Management Feature Parameter Description

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1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft B (2026-01-30)

This issue includes the following changes.

- Technical changes
None
- Editorial changes
Added descriptions of the coverage scenario configuration requirements for traditional 32T AAUs. For details, see [5.1.1.1.2 Coverage Scenarios Supported by Different RF Modules](#).
Added descriptions of the coverage scenario configuration requirements for MetaAAUs. For details, see [5.1.1.2 MetaAAUs](#).
Revised descriptions in this document.

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 07 (2025-12-14).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added support for basic beam management and 3D coverage pattern by the new module AAU5656 (2300 MHz). For details, see 5.1.1.1.2 Coverage Scenarios Supported by Different RF Modules .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added support for basic beam management and 3D coverage pattern by the new module AAU5685md (2300 MHz + 2600 MHz). For details, see 5.1.1.2 MetaAAUs .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

- Editorial changes
 - Added descriptions that a switch must be turned on before the base station reports the alarm described in [5.1.1.2 MetaAAUs](#).
 - Revised descriptions in this document.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FBFD-010015	Basic Beam Management	4 Basic Beam Management (Low-Frequency TDD)
FOFD-010100	3D Coverage Pattern	5 3D Coverage Pattern (Low-Frequency TDD)

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Basic beam management	None	4 Basic Beam Management (Low-Frequency TDD)
3D coverage pattern	None	5 3D Coverage Pattern (Low-Frequency TDD)

Differences Between High Frequency Bands and Low Frequency Bands

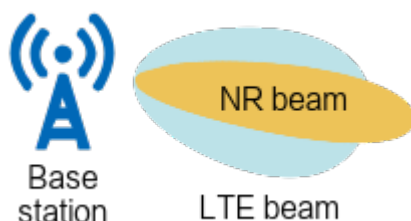
This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Basic beam management	Supported only in low frequency bands	4 Basic Beam Management (Low-Frequency TDD)
3D coverage pattern	Supported only in low frequency bands	5 3D Coverage Pattern (Low-Frequency TDD)

3 Overview

The New Radio (NR) system uses the beamforming technology to allow all types of channels and signals to use narrow beams with higher energy density and better directivity. Unlike a wide beam (such as an LTE beam), a narrow beam provides limited coverage and is unable to cover all UEs in a cell. The narrow beam also cannot direct the maximal signal energy for each UE, as shown in [Figure 3-1](#).

Figure 3-1 Comparison between wide and narrow beams



To address this issue, beam management is introduced to enable the gNodeB to manage the beams of channels and signals based on the characteristics of channels and signals and select the optimal beams for UEs. This improves coverage performance and user experience.

Beams can be divided into the following two types based on the weight policy used for beamforming:

- **Static beams:** beams generated using predefined weights. Their characteristics such as widths and directions as well as the number of beams are all fixed in a cell. The optimal beams are selected for channels and signals based on information such as cell coverage, UE distribution, and system load. This document mainly describes how to manage static beams.
- **Dynamic beams:** beams generated using weights calculated based on channel quality. Their widths and directions are dynamically adjusted based on factors such as UE positions and channel conditions. For details about how to calculate dynamic beam weights, see the downlink beamforming section in *MIMO (TDD)*.

[Table 3-1](#) describes beam classification.

Table 3-1 Beam classification

Beam Classification		Applicable Channel and Signal	Description
Static beam	Broadcast beam	Physical broadcast channel (PBCH) and synchronization signal (SS)	The PBCH and SS share the same type of beam, broadcast beam.
	Control beam	Physical uplink control channel (PUCCH), physical downlink control channel (PDCCH), channel state information-reference signal (CSI-RS), and tracking reference signal (TRS)	Each type of control channel and signal corresponds to one separate beam type. Beams for these control channels and signals are collectively referred to as control beams. Other control channels and signals are not described in this document.
Dynamic beam		Physical uplink shared channel (PUSCH) and physical downlink shared channel (PDSCH)	Each type of channel corresponds to one beam type.
Note: For details about channels and signals, see <i>Channel Management</i> .			

Static beams include broadcast beams and control beams, which are managed differently due to varying performance requirements (for example, broadcast beams have higher coverage requirements). [Table 3-2](#) details the beam management methods.

Table 3-2 Management methods of static beams

Static Beam Classification	Management Method	Section
Broadcast beam	Broadcast beams are beams for cell-level channels or signals. Various coverage scenarios are developed for broadcast beams to extend the coverage area of broadcast channels and synchronization signals in a cell, provide better cell coverage, and suit UE distribution.	For details about the default coverage scenario in TDD, see 4.1.1 Management of Broadcast Beams .
		For details about other coverage scenarios in TDD, see 5 3D Coverage Pattern (Low-Frequency TDD) .

Static Beam Classification	Management Method	Section
Control beam	Control beams are designed as required for control channels or signals to meet the coverage requirements of all UEs in a cell.	For detailed information in TDD, see 4.1.2 Management of Control Beams .

The NR system supports beam management in both high and low frequency bands. This document describes beam management only in low frequency bands. For details about beam management in high frequency bands, see *mmWave Beam Management (High-Frequency TDD)*.

In addition, emergency base stations can use "SSB coverage pattern for emergency and normal services" and "connected mode coverage pattern for emergency and normal services". These two functions adjust the coverage scopes of emergency base stations, which are often positioned at high elevations. The purpose is to reduce interference between cells served by emergency base stations and neighboring cells, while ensuring contiguous coverage by emergency base stations together with normal base stations. For details about these two functions, see *Emergency Communications Solution (Low-Frequency TDD)*.

4 Basic Beam Management (Low-Frequency TDD)

4.1 Principles

Basic beam management includes broadcast beam management and control beam management. For details, see [4.1.1 Management of Broadcast Beams](#) and [4.1.2 Management of Control Beams](#), respectively.

4.1.1 Management of Broadcast Beams

In basic beam management, broadcast beams can be managed only in the typical 3-sector networking scenario. This scenario is called the default coverage scenario in this document and specified by setting the **NRDUCellTrpBeam.CoverageScenario** parameter to **DEFAULT**.

Broadcast beams refer to synchronization signal and PBCH block (SSB) beams in an NR cell. SSB beams are cell-level beams. The gNodeB periodically sends SSB beams to broadcast synchronization signals and system information. The SSB period is specified by the **NRDUCell.SsbPeriod** parameter.

[Table 4-1](#) lists the maximum number of SSB beams in accordance with section 4.1 "Cell search" in 3GPP TS 38.213 V16.13.0.

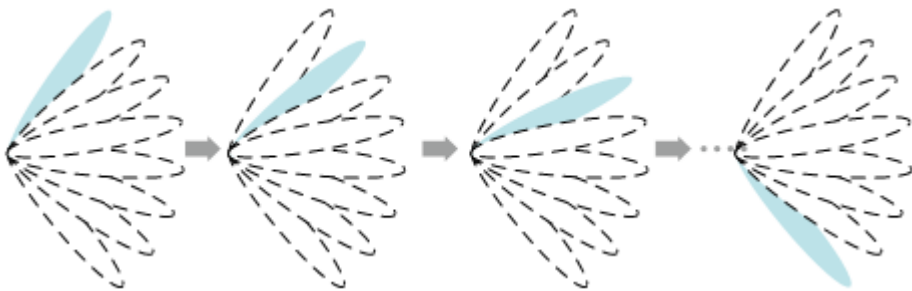
Table 4-1 Maximum number of beams on different frequencies

Working Frequency (F)	Maximum Number of Beams
1.88 GHz < F ≤ 7.125 GHz	8
F < 1.88 GHz	4
Note: Working frequencies lower than 2.3 GHz in FR1 are not supported in TDD currently.	

For an NR cell with one SSB beam, the **NRDUCell.SsbTimePos** parameter can be used to specify the SSB time-domain position. Setting different positions for different cells improves the SSB SINRs of these cells.

For an NR cell with multiple SSB beams, the beams are generally sent in different directions at different moments until the entire cell is covered, as shown in [Figure 4-1](#).

Figure 4-1 SSB beams



A UE receives signals from all beams and selects one with the best signal quality for synchronization and system information demodulation. For a non-SUL cell (with **NRDUCell.DuplexMode** not set to **CELL_SUL**) working in at least 8T mode, if the **DL_INITIAL_BEAM_SELECT_SW** option of the **NRDUCellAlgoSwitch.BeamOptSwitch** parameter is selected, downlink beams are determined based on SSB beams during initial access, increasing the access success rate; otherwise, downlink beams are determined based on PRACH beams during initial access.

4.1.1.1 TDD 32T/64T Cells

The default coverage scenario is collectively determined by the horizontal 3 dB beamwidth, vertical 3 dB beamwidth, tilt adjustment range, and azimuth adjustment range.

Different AAUs have different coverage scopes, as listed in [Table 4-2](#), due to different antenna element arrangements and hardware capabilities. The beamwidth is fixed, the tilt angle is specified by the **NRDUCellTrpBeam.Tilt** parameter, and the azimuth is specified by the **NRDUCellTrpBeam.Azimuth** parameter. The azimuth must be set to 0° in the default coverage scenario.

Table 4-2 Beamwidth, tilt, and azimuth of a TDD 32T/64T cell in the default broadcast beam coverage scenario

RF Module	Horizont al 3 dB Beamwid th	Vertical 3 dB Beamwidth	Tilt Adjustment Range	Azimuth Adjustment Range
AAU5656m (2600 MHz)	105°	4.5°	-2°~13°	0°
AAU5636m (2600 MHz)	105°	4.5°	-2°~13°	0°

RF Module	Horizontal 3 dB Beamwidth	Vertical 3 dB Beamwidth	Tilt Adjustment Range	Azimuth Adjustment Range
AAU5636 (2300 MHz)	130°	6°	-2°~13°	0°
AAU5356 (2300 MHz)	130°	6°	-2°~13°	0°
AAU5639 (2300 MHz)	130°	6°	-2°~13°	0°
AAU5368md (2300 MHz + 3500 MHz) in the 2300 MHz frequency band	105°	6°	0°~9°	0°
AAU5368md (2300 MHz + 3500 MHz) in the 3500 MHz frequency band	85°	6°	0°~9°	0°
AAU5368md (2600 MHz + 3600 MHz) in the 2600 MHz frequency band	105°	6°	0°~9°	0°
AAU5368md (2600 MHz + 3600 MHz) in the 3600 MHz frequency band	85°	6°	0°~9°	0°
Others ^a	105°	6°	-2°~13°	0°
a: "Others" refer to low-frequency TDD 32T32R and 64T64R AAUs other than the RF modules listed above.				

4.1.1.2 TDD 2T/4T/8T Cells

Table 4-3 lists the TDD 2T/4T/8T cell scenarios classified according to RF module type and antenna form.

Table 4-3 TDD 2T/4T/8T cell scenarios

RF Module Type	Antenna Form	Broadcast Beam Management Method
Book RRU	Internal antenna	Cells Served by Book RRUs with Internal Antennas

RF Module Type	Antenna Form	Broadcast Beam Management Method
Book RRU or common RRU	External antenna	Cells Served by Book RRUs or Common RRUs with External Antennas
Easy Macro AAU	Internal antenna	Cells Served by Easy Macro AAUs with Internal Antennas

Cells Served by Book RRUs with Internal Antennas

Only the default coverage scenario is supported, that is, the **NRDUCellTrpBeam.CoverageScenario** parameter can only be set to **DEFAULT**. [Table 4-4](#) describes the broadcast beam management methods.

Table 4-4 Broadcast beam management methods for cells served by Book RRUs with internal antennas

RF Module	TX/RX Mode	Broadcast Beam Management Method
RRU5836E	4T4R	Two SSB beams are generally sent in a cell. If broadcast beam densification is enabled by selecting the SCENARIO_BEAM_DENSIFY_ALGO_SW option of the NRDUCellTrpBeam.ScenarioBeamAlgoSw parameter, broadcast beams are sent based on the maximum number of beams supported under the current slot configuration. This improves cell coverage but consumes more time-frequency resources.

Cells Served by Book RRUs or Common RRUs with External Antennas

The default coverage scenario and some 3D coverage scenarios are supported. [Table 4-5](#) describes the broadcast beam management methods in the default coverage scenario. For details about 3D coverage scenarios, see [5.1.2 TDD 8T Cells](#).

Table 4-5 Broadcast beam management methods for cells served by Book RRUs or common RRUs with external antennas

TX/RX Mode	Broadcast Beam Management Method
1T1R	One SSB beam is sent in a cell.

TX/RX Mode	Broadcast Beam Management Method
2T2R	Depending on the NRDUCellTrp.AntPolarization parameter setting: • If this parameter is set to DEFAULT, SSB beams are sent on two RF channels in turn in a cell. There are two SSB beams in total. • If this parameter is set to SINGLE_POLARIZATION, SSB beams are sent in a cell based on the maximum number of SSBs supported under the current slot configuration. This consumes more time-frequency resources. The coverage of broadcast channels and synchronization signals increases by 1 dB to 3 dB (the power of data channels remains unchanged). This may absorb UEs at the cell edge, causing KPIs to deteriorate. • If this parameter is set to DOUBLE_POLARIZATION, one SSB beam is sent simultaneously on two RF channels. There is one SSB beam in total.
4T4R	Depending on the NRDUCellTrp.AntPolarization parameter setting: • If this parameter is set to DEFAULT, SSB beams are sent on four RF channels in turn in a cell. There are four SSB beams in total. • If this parameter is set to SINGLE_POLARIZATION or DOUBLE_POLARIZATION, SSB beams are sent in a cell based on the maximum number of SSBs supported under the current slot configuration. This consumes more time-frequency resources. The coverage of broadcast channels and synchronization signals increases by 3 dB to 6 dB (the power of data channels remains unchanged). This may absorb UEs at the cell edge, causing KPIs to deteriorate.

TX/RX Mode	Broadcast Beam Management Method
8T8R ^a	Depending on the NRDUCellTrp.AntPolarization parameter setting: • If this parameter is set to DEFAULT or DOUBLE_POLARIZATION, the base station selects weights from a 4H1V antenna weight file. Taking the ATD4516R8 antenna as an example, the horizontal 3 dB beamwidth is 90° and the vertical 3 dB beamwidth is 6°. SSB beams are sent on eight RF channels of the cell at the same time, and there are a total of four SSB beams. • If this parameter is set to 2H2V_DOUBLE_POLARIZATION, the base station selects weights from a 2H2V antenna weight file. If broadcast beam densification is enabled by selecting the SCENARIO_BEAM_DENSIFY_ALGO_SW option of the NRDUCellTrpBeam.ScenarioBeamAlgoSw parameter, broadcast beams are sent based on the maximum number of beams supported under the current slot configuration. This improves cell coverage but consumes more time-frequency resources.
Note: a: In this document, the beam pattern of the 8T8R RRU configured with an 8T8R antenna (for example, a 4H1V ATD4516R8 antenna) is used as an example. Huawei-manufactured antennas or antennas that comply with the Next Generation Mobile Network (NGMN) standard are recommended. If the antennas of other vendors or the antennas that do not comply with the NGMN standard are used, the function may fail to take effect, affecting the coverage area of the cell. In addition, the tilt adjustment range is determined by external antennas.	

Cells Served by Easy Macro AAUs with Internal Antennas

The default coverage scenario and some 3D coverage scenarios are supported. [Table 4-6](#) describes the broadcast beam management methods in the default coverage scenario (that is, when **NRDUCellTrpBeam.CoverageScenario** is set to **DEFAULT**). For details about 3D coverage scenarios, see [5.1.2 TDD 8T Cells](#).

Table 4-6 Broadcast beam management methods for cells served by Easy Macro AAUs with internal antennas

RF Module	TX/RX Mode	Broadcast Beam Management Method
AAU5243	8T8R	In narrow beam mode, four SSB beams are sent in a cell. In wide beam mode, two SSB beams are sent in a cell. If broadcast beam densification is enabled by selecting the SCENARIO_BEAM_DENSIFY_ALGO_SW option of the NRDUCellTrpBeam.ScenarioBeamAlgoSw parameter, broadcast beams are sent in a cell based on the maximum number of beams supported under the current slot configuration, regardless of whether the narrow beam mode or wide beam mode is used. This improves cell coverage but consumes more time-frequency resources.
AAU5246	8T8R	In narrow beam mode, four SSB beams are sent in a cell. The RVDSUBUNIT.VBEAMWIDTH parameter must be set to 200 , which means that the vertical 3 dB beamwidth is 21°. In wide beam mode, four SSB beams are sent in a cell. The RVDSUBUNIT.VBEAMWIDTH parameter must be set to 400 , which means that the vertical 3 dB beamwidth is 43°. If broadcast beam densification is enabled by selecting the SCENARIO_BEAM_DENSIFY_ALGO_SW option of the NRDUCellTrpBeam.ScenarioBeamAlgoSw parameter, broadcast beams are sent in a cell based on the maximum number of beams supported under the current slot configuration, regardless of whether the narrow beam mode or wide beam mode is used. This improves cell coverage but consumes more time-frequency resources.

4.1.2 Management of Control Beams

Management of control beams refers to managing the beams of control channels and reference signals such as PUCCH, PDCCH, and CSI-RS.

Beam Management for Uplink Channels

For the PUCCH, beams with the largest RSRP are selected based on SRS beam measurement.

Beam Management for Downlink Channels and Reference Signals

For the PDCCH:

- Common PDCCH: Transmission beams are consistent with SSB beams.
- UE-specific PDCCH: Beams with the largest SRS RSRP are selected as transmission beams.

PDCCH precoding enhancement

For the PDCCH in low-frequency TDD cells with at least 2T2R, CSI-RS-based PMIs can be used in PDCCH beamforming to generate better beams and improve coverage and capacity. This function can be enabled by setting the **NRDUCellPdcchAlgo.PdcchPrecodeEnhPolicy** parameter to **ENH_PMI**. If this function is enabled, it takes effect for a UE when the UE SRS SINR before equalization is less than the value of **NRDUCellPdcchAlgo.PdcchPrecodeEnhPreSinrThld**; it does not take effect for the UE in other cases.

PDCCH beam robustness

UEs may fail to detect the PDCCH in two scenarios: (1) PDCCH transmission beams are selected based on inaccurate SRS beam measurements (which aim to find optimal SRS beams) due to interference. (2) PDCCH beamforming relies on abnormal CSI-RS measurements reported by UEs. The PDCCH beam robustness function addresses these issues by switching the PDCCH beam selection method between SRS beam measurement and CSI-RS measurement to reduce missing PDCCH detections.

If the number of consecutive uplink or downlink scheduling results indicating discontinuous transmission (DTX) for a UE meets or exceeds a threshold, the gNodeB determines that the UE has consecutively missed the PDCCH and triggers PDCCH beam selection method switching for the UE. The DTX number threshold is specified by the **NRDUCellPdcchAlgo.PdcchBeamRobDtxThld** parameter. If this parameter is set to **0**, PDCCH beam robustness does not take effect. This function is controlled by the **PDCCH_BEAM_ROBUST_SW** option of the **NRDUCellPdcch.PdcchAlgoSwitch** parameter.

SRS beam selection optimization

When there is SRS interference, a beam may not be optimal even if it has the largest SRS RSRP based on antenna-selection SRS measurements. The SRS beam selection optimization function allows the gNodeB to use non-antenna-selection SRS measurements to select beams with the largest SRS RSRP for PDCCH and CSI-RS transmission. This function is controlled by the **SRS_BEAM_SELECT_OPT_SW** option of the **NRDUCellSrs.SrsAlgoSwitch** parameter.

For the CSI-RS, beams with the largest SRS RSRP are selected as transmission beams. For low-frequency TDD cells with at least 32T32R, different CSI-RS beam types and quantities can be configured for CSI reporting and PDSCH PMI-based weight calculation.

- The beam type is specified by the **NRDUCellCsirs.CsirsBeamType** parameter.
- The beam quantity is specified by the **NRDUCellCsirs.CsirsCellResourceNum** parameter. Using more CSI-RS beams helps enhance the CSI-RS coverage and improve the beam direction accuracy, but may increase the chances of beam switching. The **NRDUCellCsirs.CsirsBeamSwitchThld** parameter determines

the probability of beam switching. For details about CSI-RS, see section 6.3.2 "Radio resource control information elements" in 3GPP TS 38.331 V17.3.0 and section 7.4.1.5 "CSI reference signals" in 3GPP TS 38.211 V17.4.0.

4.2 Network Analysis

4.2.1 Benefits

Broadcast Beam Management

Broadcast beam management uses beam sweeping to direct the transmit energy at target UEs by selecting the optimal beams for UEs. This increases the SNR for demodulation by target UEs and improves the transmission success rate, and also extends the coverage area of the broadcast channels and synchronization signals, thereby increasing the cell coverage radius.

Control Beam Management

Control beam management improves the beam directivity accuracy by selecting the optimal beams for UEs, thereby increasing the SNR for demodulation by target UEs, improving the transmission success rate, and extending the coverage area of the UE-level channels and signals.

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

This function has no impact on the network.

4.3 Requirements

4.3.1 Licenses

None

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

None

4.3.2.2 Mutually Exclusive Functions

Function Name	Function Switch	Reference	Description
SCell CSI-RS beam sweeping	CARR_CSI_BEAM_SWEEPING_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoSwitch parameter	<i>Carrier Aggregation</i>	SCell CSI-RS beam sweeping must be disabled when the NRDUCellCsirs.CsirsCellResourceNum parameter is set to 1_RESOURCE for a cell and any of the following conditions is not met: - The NRDUCellTrp.TxRxMode parameter is set to 64T64R or 128T128R in at least one NRDUCellTrp MO associated with this cell through the NRDUCellCsirs.NrDuCellId parameter. - The NRDUCellCsirs.FR1MaxCellCsirsPortNum parameter is set to 32PORT.
Connected mode coverage pattern for emergency and normal services	NRDUCellTrpBeam.ConnModeCovScenarioExt	<i>Emergency Communications Solution (Low-Frequency TDD)</i>	If the NRDUCellCsirs.CsirsBeamType parameter is set to TYPE0, the NRDUCellCsirs.FR1MaxCellCsirsPortNum parameter is set to 16PORT, and the NRDUCellCsirs.CsirsCellResourceNum parameter is set to 1_RESOURCE for a cell, then the NRDUCellTrpBeam.ConnModeCovScenarioExt parameter must be set to DEFAULT for the cell.

4.3.2.3 Function Impacts

Function Name	Function Switch	Reference	Description
RF channel intelligent shutdown	RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	You are advised to select the DL_INITIAL_BEAM_SELECT_SW option of the NRDUCellAlgoSwitch.BeamOptSwitch parameter before enabling RF channel intelligent shutdown. Otherwise, the RRC connection setup success rate decreases.

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.3.4 Cells

None

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

Table 4-7 describes the parameters used for function activation.

Table 4-7 Parameters used for activation for a TDD cell

Parameter Name	Parameter ID	Setting Notes
Coverage Scenario	NRDUCellTrpBeam.CoverageScenario	This parameter must be set to DEFAULT .
Tilt	NRDUCellTrpBeam.Tilt	Set this parameter based on the network plan. If this parameter is set to the default value 255 in scenarios with 32T32R or higher, the tilt is 6°. The antenna tilt is the value specified by this parameter plus the mechanical tilt. For RF modules that support the RET system, tilt adjustment is implemented by phase shifters. Changing the value of this parameter will affect the tilts of all channels and signals. All 32T32R modules support the RET system. For details, see section "AAU RET System" in the related hardware description. No 64T64R modules support the RET system.
Azimuth	NRDUCellTrpBeam.Azimuth	Set this parameter based on the network plan.
SSB Period	NRDUCell.SsbPeriod	Retain the default value.
SSB Time Position	NRDUCell.SsbTimePos	Retain the default value.

Parameter Name	Parameter ID	Setting Notes
Antenna Polarization	NRDUCellTrp.AntPolarization	This parameter takes effect only in 2T, 4T, or 8T cells. It is recommended that the parameter be set based on the antenna polarization mode to extend the coverage area of broadcast channels and synchronization signals.
Beam Optimization Switch	NRDUCellAlgoSwitch.BeamOptSwitch	This parameter takes effect only in cells with at least 8T. It is recommended that the DL_INITIAL_BEAM_SELECT_SW option be selected for low-speed cells. During initial access, using SSB beams provides a higher access success rate than using PRACH beams. It is recommended that this option be deselected for high-speed cells, so that PRACH beams are used during initial access.
Scenario-based Beam Algorithm Switch	NRDUCellTrpBeam.ScenarioBeamAlgoSw	If coverage scenarios are different among neighboring cells, the cells will interfere with each other and downlink throughput will decrease. It is recommended that the SCENARIO_BEAM_DENSIFY_ALGO_SW option be selected to reduce interference.
CSI-RS Beam Type	NRDUCellCsirs.CsirsBeamType	Set this parameter based on the network plan. The value TYPE0 applies to horizontal open areas, TYPE1 applies to densely populated high-rise buildings, and TYPE2 applies to common urban areas.

Parameter Name	Parameter ID	Setting Notes
CSIRS Cell Resource Number	NRDUCellCsirs.CsirsCellResourceNum	The default value is recommended. The value FD_RESOURCE is not recommended in the following scenarios: - Highway and high-speed railway scenarios. If the value is used in such scenarios, more beam switching will occur. - Scenarios with an extremely heavy load. If the value is used in such scenarios, more CSI-RS resources will be required.
CSIRS Beam Switch Thld	NRDUCellCsirs.CsirsBeamSwitchThld	Retain the default value.
PDCCH Precode Enh Policy	NRDUCellPdcchAlgo.PdcchPrecodeEnhPolicy	The value ENH_PMI is recommended.
PDCCH Precode Enh PreSINR Thld	NRDUCellPdcchAlgo.PdcchPrecodeEnhPreSnrThld	It is recommended that this parameter be set to -10 in the following scenarios: - The NRDUCellCsirs.CsirsCellResourceNum parameter is set to 1_RESOURCE, 2_RESOURCE, or 4_RESOURCE for a 64T cell. - The NRDUCellCsirs.CsirsCellResourceNum parameter is set to 1_RESOURCE or 2_RESOURCE for a 32T cell. In other scenarios, this parameter can be set to 21 to achieve higher gains.
PDCCH Algorithm Switch	NRDUCellPdcch.PdcchAlgoSwitch	It is recommended that the PDCCH_BEAM_ROBUST_SW option be selected if the PDCCH beam selection method needs to be switched through the PDCCH beam robustness function.

Parameter Name	Parameter ID	Setting Notes
PDCCH Beam Robust DTX Threshold	NRDUCellPdcchAlgo.PdcchBeamRobDtxThld	The value 4 is recommended.
SRS Algorithm Switch	NRDUCellSrs.SrsAlgoSwitch	If the beam with the largest SRS RSRP based on antenna-selection SRS measurements is not optimal, it is recommended that the SRS_BEAM_SELECT_OPT_SW option be selected to use non-antenna-selection SRS measurements to select beams with the largest SRS RSRP for PDCCH and CSI-RS transmission.

NOTICE

- It is recommended that the **NRDUCellTrpBeam.CoverageScenario** parameter or the **SCENARIO_BEAM_DENSIFY_ALGO_SW** option of the **NRDUCellTrpBeam.ScenarioBeamAlgoSw** parameter be adjusted during off-peak hours. This is because the service performance of UEs in connected mode or the access performance will be affected within 5 minutes after the adjustment. The related indicators include the UE throughput, access KPIs, handover KPIs, and service drop KPIs.
- Changing the value of the **NRDUCellTrp.AntPolarization** parameter will cause a cell to restart. Therefore, perform this operation during off-peak hours.

4.4.1.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

- 32T/64T cells

```
//Setting the coverage scenario and tilt of broadcast beams
MOD NRDUCELLTRPBEAM: NrDuCellTrpId=1, CoverageScenario=DEFAULT, Tilt=255, Azimuth=0;
//Setting SSB parameters for the cell
```

```

MOD NRDUCELL: NrDuCellId=1, DuplexMode=CELL_TDD, SsbPeriod=MS20, SsbTimePos=DEFAULT;
//Setting CSI-RS beams for the cell
MOD NRDUCELLCSIRS: NrDuCellId=1, CsiPeriod=SLOT40, CsirsCellResourceNum=4_RESOURCE,
CsirsBeamSwitchThld=2, CsirsBeamType=TYPE0;
//Setting the PDCCH precoding enhancement policy
MOD NRDUCELLPDCCHALGO: NrDuCellId=1, PdcchPrecodingEnhPolicy=ENH_PMI,
PdcchPrecodingEnhPreSnrThld=-10;
//Enabling initial downlink beam selection
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, BeamOptSwitch=DL_INITIAL_BEAM_SELECT_SW-1;
//(Optional, recommended) Enabling PDCCH beam robustness when the PDCCH beam selection
method needs to be switched
MOD NRDUCELLPDCCH: NrDuCellId=1, PdcchAlgoSwitch=PDCCH_BEAM_ROBUST_SW-1;
MOD NRDUCELLPDCCHALGO: NrDuCellId=1, PdcchBeamRobDtxThld=4;
//(Optional, recommended) Enabling SRS beam selection optimization if the beam with the largest
SRS RSRP based on antenna-selection SRS measurements is not optimal
MOD NRDUCELLSRS: NrDuCellId=1, SrsAlgoSwitch=SRS_BEAM_SELECT_OPT_SW-1;

```

- **2T/4T/8T cells**

```

//(Optional, recommended) Enabling PDCCH beam robustness when the PDCCH beam selection
method needs to be switched
MOD NRDUCELLPDCCH: NrDuCellId=1, PdcchAlgoSwitch=PDCCH_BEAM_ROBUST_SW-1;
MOD NRDUCELLPDCCHALGO: NrDuCellId=1, PdcchBeamRobDtxThld=4;
//(Optional, recommended) Enabling SRS beam selection optimization if the beam with the largest
SRS RSRP based on antenna-selection SRS measurements is not optimal
MOD NRDUCELLSRS: NrDuCellId=1, SrsAlgoSwitch=SRS_BEAM_SELECT_OPT_SW-1;

```

Cells served by Book RRUs with internal antennas

```

//Setting the coverage scenario of broadcast beams
MOD NRDUCELLTRPBEAM: NrDuCellTrpId=1, CoverageScenario=DEFAULT;
//Enabling broadcast beam densification
MOD NRDUCELLTRPBEAM: NrDuCellTrpId=1, CoverageScenario=DEFAULT,
ScenarioBeamAlgoSw=SCENARIO_BEAM_DENSIFY_ALGO_SW-1;

```

Cells served by Book RRUs or common RRUs with external antennas

```

//Setting the coverage scenario of broadcast beams
MOD NRDUCELLTRPBEAM: NrDuCellTrpId=1, CoverageScenario=DEFAULT;
//Setting the antenna polarization mode (using 2T2R as an example)
MOD NRDUCELLTRP: NrDuCellTrpId=1, TxRxMode=2T2R, AntPolarization=DEFAULT;
//Enabling broadcast beam densification
MOD NRDUCELLTRPBEAM: NrDuCellTrpId=1, CoverageScenario=DEFAULT,
ScenarioBeamAlgoSw=SCENARIO_BEAM_DENSIFY_ALGO_SW-1;

```

Cells served by Easy Macro AAUs with internal antennas

AAU5243

```

//Setting the coverage scenario of broadcast beams
MOD NRDUCELLTRPBEAM: NrDuCellTrpId=1, CoverageScenario=DEFAULT;
//Setting the narrow beam mode
MOD RETSUBUNIT: DEVICENO=1, SUBUNITNO=1, TILT=30;
MOD RETSUBUNIT: DEVICENO=2, SUBUNITNO=1, TILT=30;
//Enabling broadcast beam densification
MOD NRDUCELLTRPBEAM: NrDuCellTrpId=1, CoverageScenario=DEFAULT,
ScenarioBeamAlgoSw=SCENARIO_BEAM_DENSIFY_ALGO_SW-1;

```

AAU5246

```

//Setting the coverage scenario of broadcast beams
MOD NRDUCELLTRPBEAM: NrDuCellTrpId=1, CoverageScenario=DEFAULT;
//Setting the narrow beam mode
MOD RVDSUBUNIT: DEVICENO=0, SUBUNITNO=1, VBEAMWIDTH=200;
//Enabling broadcast beam densification
MOD NRDUCELLTRPBEAM: NrDuCellTrpId=1, CoverageScenario=DEFAULT,
ScenarioBeamAlgoSw=SCENARIO_BEAM_DENSIFY_ALGO_SW-1;

```

Deactivation Command Examples

Deactivation is not required for this function. You can determine whether to restore the settings of other parameters based on network conditions.

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

This function is a basic function and takes effect by default. Activation verification is not required.

For a cell with at least 32T32R, you can run the **LST NRDUCELLTRPSUBBEAM** command to query information about each SSB beam in the cell or use the counters listed in [Table 4-8](#) to observe the number of UEs and traffic volume under the optimal SSB beam.

Table 4-8 Number of UEs and traffic volume under the optimal SSB beam

Counter ID	Counter Name	Description
1911827216	N.User.OptimalSSBBeam.Avg	Average number of UEs under the optimal SSB beam
1911827224	N.MAC.ThpVol.UL.OptimalSSBBeam	Total uplink traffic volume at the MAC layer under the optimal SSB beam
1911827225	N.MAC.ThpVol.DL.OptimalSSBBeam	Total downlink traffic volume at the MAC layer under the optimal SSB beam

 NOTE

The optimal SSB beam of UEs is determined based on SRS measurement and may not be the SSB beam actually used by the UEs for access. The measured number of UEs and traffic volume under some SSB beams may be 0, which does not indicate the actual coverage of these SSB beams.

4.4.3 Network Monitoring

This function is a basic function that ensures basic service performance. Monitoring is not required.

5 3D Coverage Pattern (Low-Frequency TDD)

5.1 Principles

The 3D coverage pattern function applies only to broadcast beams. It enhances the coverage of broadcast channels and synchronization signals in different scenarios and better matches the cell coverage scope and UE distribution. The gNodeB supports multiple beam coverage scenarios.

Operators can select beam coverage scenarios based on their requirements to solve the problems of limited cell coverage and neighboring cell interference in different networking scenarios.

The 3D coverage pattern function allows cells with at least 8T to provide different 3D coverage in different networking scenarios. For details, see [5.1.1 TDD 32T/64T Cells](#) and [5.1.2 TDD 8T Cells](#).

5.1.1 TDD 32T/64T Cells

The 3D coverage scenario of a cell is specified by the **NRDUCellTrpBeam.CoverageScenario** parameter.

A coverage scenario is collectively determined by the horizontal 3 dB beamwidth, vertical 3 dB beamwidth, tilt adjustment range, and azimuth adjustment range. The beamwidth is not adjustable, the tilt is specified by the **NRDUCellTrpBeam.Tilt** parameter, and the azimuth is specified by the **NRDUCellTrpBeam.Azimuth** parameter. By adjusting the tilt and azimuth, you can change the coverage radius, coverage shape, and beam direction to meet different coverage requirements and achieve flexible networking.

Traditional AAUs and MetaAAUs have different antenna array structures and support different coverage scenarios. For details, see [5.1.1.1 Traditional AAUs](#) and [5.1.1.2 MetaAAUs](#), respectively.

5.1.1.1 Traditional AAUs

5.1.1.1.1 3D Coverage Scenarios

Table 5-1 lists the 3D coverage scenarios supported by traditional AAUs.

Table 5-1 3D coverage scenarios supported by traditional AAUs

Scenario ID	Horizontal 3 dB Beamwidth	Vertical 3 dB Beamwidth	Tilt Adjustment Range	Azimuth Adjustment Range	Description
SCENARIO_1	110°	6°	-2°~13°	0°	The horizontal coverage indicated by the horizontal 3 dB beamwidth decreases with the increase of the scenario ID. The vertical coverage indicated by the vertical 3 dB beamwidth (6°) applies to lower-floor coverage.
SCENARIO_2	90°	6°	-2°~13°	-10°~10°	
SCENARIO_3	65°	6°	-2°~13°	-22°~22°	
SCENARIO_4	45°	6°	-2°~13°	-32°~32°	
SCENARIO_5	25°	6°	-2°~13°	-42°~42°	

Scenario ID	Horizontal 3 dB Beamwidth	Vertical 3 dB Beamwidth	Tilt Adjustment Range	Azimuth Adjustment Range	Description
SCENARIO_6	110°	12°	0°~9°	0°	The horizontal coverage indicated by the horizontal 3 dB beamwidth decreases with the increase of the scenario ID. The vertical coverage indicated by the vertical 3 dB beamwidth (12°) applies to middle-floor coverage.
SCENARIO_7	90°	12°	0°~9°	-10°~10°	
SCENARIO_8	65°	12°	0°~9°	-22°~22°	
SCENARIO_9	45°	12°	0°~9°	-32°~32°	
SCENARIO_10	25°	12°	0°~9°	-42°~42°	
SCENARIO_11	15°	12°	0°~9°	-47°~47°	

Scenario ID	Horizontal 3 dB Beamwidth	Vertical 3 dB Beamwidth	Tilt Adjustment Range	Azimuth Adjustment Range	Description
SCENARIO_12	110°	- AAU5356 (2600 MHz): 18° - AAU5356 (3500 MHz/3700 MHz): 15° - Others: 25°	6°	0°	The horizontal coverage indicated by the horizontal 3 dB beamwidth decreases with the increase of the scenario ID. The vertical coverage indicated by the vertical 3 dB beamwidth (25°) applies to upper-floor coverage.
SCENARIO_13	65°	- AAU5356 (2600 MHz): 18° - AAU5356 (3500 MHz/3700 MHz): 15° - Others: 25°	6°	-22°~22°	
SCENARIO_14	45°	- AAU5356 (2600 MHz): 18° - AAU5356 (3500 MHz/3700 MHz): 15° - Others: 25°	6°	-32°~32°	

Scenario ID	Horizontal 3 dB Beamwidth	Vertical 3 dB Beamwidth	Tilt Adjustment Range	Azimuth Adjustment Range	Description
SCENARIO_15	25°	- AAU5356 (2600 MHz): 18° - AAU5356 (3500 MHz/3700 MHz): 15° - Others: 25°	6°	-42°~42°	
SCENARIO_16	15°	25°	6°	-47°~47°	
SCENARIO_33	65°	12°	0°~9°	-22°~22°	This scenario is supported only by emergency base stations and is recommended during emergencies.
SCENARIO_34	105°	6°	-2°~13°	0°	This scenario is supported only by emergency base stations and is recommended in normal situations.

Scenario ID	Horizontal 3 dB Beamwidth	Vertical 3 dB Beamwidth	Tilt Adjustment Range	Azimuth Adjustment Range	Description
SCENARIO_35	65°	6°	-2°~13°	-22°~22°	This scenario is supported only by emergency base stations and is recommended in normal situations.
CUSTOMIZED_BEAM_SCENARIO	15° or 30° for a single beam	6°	-2°~13°	-47°~47°	Perform configuration under the guidance of Huawei network optimization and planning engineers.

NOTE

- Assume that the beam transmit direction perpendicular to the front of the AAU is defined as the normal direction (0°). Then, a positive tilt adjustment value indicates downward adjustment, a negative tilt adjustment value indicates upward adjustment, a positive azimuth adjustment value indicates rightward adjustment, and a negative azimuth adjustment value indicates leftward adjustment.
- The horizontal 3 dB beamwidth of a single beam is 17° or 35° when the AAU5639 (2300 MHz) works in the CUSTOMIZED_BEAM_SCENARIO.

Example of Coverage Scenario Selection

Operators of live networks need to determine coverage scenarios based on factors such as site height, inter-site distance, UE distribution, coverage scope, and RF module capabilities.

- If there are high requirements for horizontal coverage, you are advised to configure SCENARIO_1, SCENARIO_6, or SCENARIO_12 to improve the horizontal beam gain and coverage at the cell edge.
- If there are fixed interference sources at the cell edge, you are advised to configure SCENARIO_2, SCENARIO_3, SCENARIO_7, SCENARIO_8, or SCENARIO_13 to narrow down the horizontal coverage and reduce the interference.

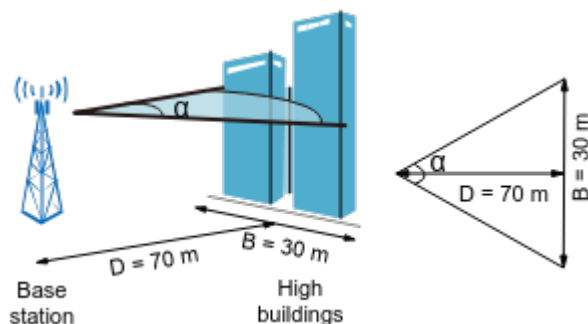
- If there is only an isolated building, you are advised to configure SCENARIO_4, SCENARIO_5, SCENARIO_9, SCENARIO_10, SCENARIO_11, SCENARIO_14, SCENARIO_15, or SCENARIO_16 for small horizontal coverage.

The following is an example of how to calculate and select a coverage scenario. Assume that the horizontal 3 dB beamwidth is indicated by α in Figure 5-1 and the vertical 3 dB beamwidth is indicated by β in Figure 5-2. They together determine the coverage scenario. In the figures, B indicates the horizontal width of the buildings, D indicates the perpendicular distance between the base station and the buildings, h indicates the height of the base station, and H indicates the height of the buildings.

Step 1 Calculate the horizontal 3 dB beamwidth.

If B is 30 m and D is 70 m, then α is 24° , as shown in Figure 5-1. According to Table 5-1, the appropriate scenarios are SCENARIO_5, SCENARIO_10, and SCENARIO_15.

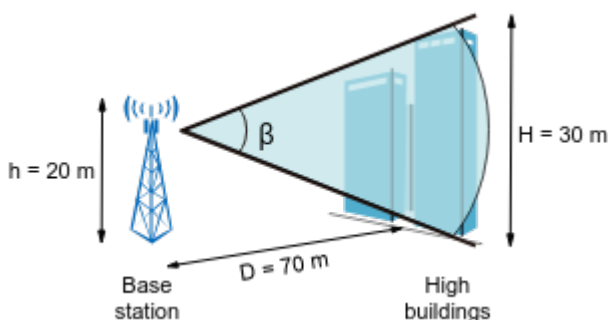
Figure 5-1 Calculating the horizontal 3 dB beamwidth



Step 2 Calculate the vertical 3 dB beamwidth.

If D is 70 m, h is 20 m, and H is 30 m, then β is 23° . According to Table 5-1, the appropriate scenarios are SCENARIO_12 to SCENARIO_16.

Figure 5-2 Calculating the vertical 3 dB beamwidth



Step 3 Select SCENARIO_15 as the coverage scenario as it is in the intersection of scenarios determined in Step 1 and Step 2, meeting both horizontal and vertical coverage requirements.

----End

Customized Coverage Scenarios

If coverage scenarios SCENARIO_1 to SCENARIO_16 do not meet actual requirements, broadcast beam coverage scenarios can be customized. Specifically, the following parameters can be customized: the number of beams, tilt, azimuth, beamwidth, and power offset.

For customized broadcast beam coverage scenarios, information about each beam must be configured first, and then the **NRDUCellTrpBeam.CoverageScenario** parameter must be set to **CUSTOMIZED_BEAM_SCENARIO** to enable the scenarios. Specifically,

- The number of beams can be customized. The ID of each beam is specified by the **NRDUCellTrpCustBeam.BeamId** parameter. [Table 5-2](#) lists the maximum number of beams supported by different slot configurations. In customized coverage scenarios, beam IDs must start with 0 and be consecutive.

Table 5-2 Maximum number of beams supported by different slot configurations

NRDUCell.SlotAssignment	Number of GP Symbols in a Self-Contained Slot	Maximum Number of Beams in a Customized Coverage Scenario
4_1_DDDSU	1~6	7
	18	3
8_2_DDDDDDDSUU	1~4	8
	5~6	7
	18	6
8_2_DDDSUUDDDD	1~4	6
	5~6	4
	18	3

For details about slot configuration and the number of GP symbols in a self-contained slot, see *Standards Compliance*.

- The tilt and azimuth of each beam can be customized using the **NRDUCellTrpCustBeam.Tilt** and **NRDUCellTrpCustBeam.Azimuth** parameters.
- The horizontal 3 dB beamwidth of each beam can be customized using the **NRDUCellTrpCustBeam.SsbBeamHorizWidth** parameter. If this parameter is set to **DEFAULT**, the horizontal beamwidth of a single beam is 15°. If this parameter is set to **WIDTH_30**, the horizontal beamwidth of a single beam is 30°.
- The power offset of each beam can be customized using the **NRDUCellTrpCustBeam.MaxSsbPwrOffset** parameter. When the horizontal

beamwidth of a single beam is set to 30°, this parameter can be set to 3 dB as required to improve the beam coverage performance.

NOTE

In customized coverage scenarios, the **NRDUCellTrpBeam.Tilt** parameter must be set to 255, and the **NRDUCellTrpBeam.Azimuth** and **NRDUCellTrpBeam.MaxSsbPwrOffset** parameters must be set to 0.

5.1.1.1.2 Coverage Scenarios Supported by Different RF Modules

Table 5-3 lists the coverage scenarios supported by different RF modules.

Table 5-3 Coverage scenarios supported by traditional AAUs

RF Module	TX/RX Mode	Supported Coverage Scenarios
AAU5310 (3700 MHz) AAU5313 (3500 MHz or 3700 MHz) AAU5339 (3500 MHz or 3700 MHz) AAU5339w (3500 MHz or 3700 MHz) AAU5811 (3700 MHz) AAU5831 (3500 MHz or 3700 MHz) AAU5336 (2300 MHz) AAU5349 (3600 MHz) AAU5336w (2600 MHz or 3500 MHz) AAU5356 (2300 MHz, 2600 MHz, 3500 MHz, or 3700 MHz) AAU5366 (3600 MHz) AAU5832f (3700 MHz) AAU5368 (3600 MHz) AAU5832a (3500 MHz)	32T32R	SCENARIO_1, SCENARIO_6~SCENARIO_8, SCENARIO_12~SCENARIO_15, and CUSTOMIZED_BEAM_SCENARIO

RF Module	TX/RX Mode	Supported Coverage Scenarios
AAU5649 (3600 MHz) AAU5639w (3500 MHz, 3600 MHz, or 3700 MHz) AAU5639 (2300 MHz, 3500 MHz, or 3700 MHz) AAU5636w (3500 MHz or 3700 MHz) AAU5636 (2300 MHz or 3400 MHz) AAU5631 (2600 MHz) AAU5613 (3500 MHz or 3700 MHz) AAU5835h (3600 MHz) AAU5835f (3400 MHz, 3500 MHz, 3600 MHz, or 3700 MHz) AAU5833f (3500 MHz or 3700 MHz) AAU5825g (2600 MHz) AAU5825a (2300 MHz or 2600 MHz) AAU5827a (2600 MHz) AAU5826g (2600 MHz) AAU5646 (3600 MHz) AAU5656 (2300 MHz, 2600 MHz, 3500 MHz, or 3700 MHz) AAU5836f (3500 MHz)	64T64R	SCENARIO_1~SCENARIO_16, and CUSTOMIZED_BEAM_SCENARIO
AAU5614 (2600 MHz) ^a AAU5636 (2600 MHz) AAU5639w (2600 MHz) AAU5636w (2600 MHz)	64T64R	SCENARIO_1~SCENARIO_16, SCENARIO_33~SCENARIO_35, and CUSTOMIZED_BEAM_SCENARIO
a: If any of coverage scenarios SCENARIO_3 to SCENARIO_5 is used for the AAU5614 (2600 MHz), the SCENARIO_BEAM_OPT_SW option of the NRDUCellBeamAlgo.WeightAlgoSwitch parameter must be selected.		

The configuration requirements for coverage scenarios are as follows:

- Different AAUs support different coverage scenarios, which can be queried by running the **DSP NRDUCELLTRP** command. The base station reports ALM-29871 NR DU Cell TRP Capability Degraded, with the **Specific Problem** being **Beam configuration error** (that is, the coverage scenario, tilt angle, or azimuth exceeds the capability of the RF module) in the following situations:
 - An unsupported coverage scenario is configured. If it is configured for a cell before the cell is set up, the default coverage scenario will take effect for the cell. If the coverage scenario of a cell is changed to an unsupported one when the cell is in the active state, the change will not

take effect and the last valid value will be retained. If the coverage scenario configured for a cell is not supported by a new AAU that replaces an old one, the default coverage scenario will take effect for the cell.

- An unsupported tilt or azimuth is configured. If it is configured for a cell before the cell is set up, the valid tilt or azimuth closest to the configured value will take effect for the cell. (If multiple values are closest to the configured one, the minimum value will take effect.) If the tilt or azimuth of a cell is changed to an unsupported one when the cell is in the active state, the change will not take effect and the last valid value will be retained. If the tilt or azimuth configured for a cell is not supported by a new AAU that replaces an old one, the valid tilt or azimuth closest to the configured value will take effect for the cell. (If multiple values are closest to the configured one, the minimum value will take effect.)
- When multiple cells are configured in the same frequency band on a 32T32R RF module, these cells must meet the following requirements. Otherwise, cell coverage performance is affected, and if the **CARR_BEAM_CONFIG_DIFF_ALM_SW** option of the **gNBOamParam.AlarmSwitch** parameter is selected, the base station will report **ALM-29871 NR DU Cell TRP Capability Degraded** with the alarm cause being **Inconsistent beam configurations between co-sector carriers**.
 - The **NRDUCellTrpBeam.Tilt** parameter values of all cells must be the same.
 - The coverage scenarios of multiple NR cells must meet the requirements in [Table 5-4](#).

Table 5-4 Requirements for the coverage scenarios of multiple NR cells

Coverage Scenario of One NR Cell	Coverage Scenarios of Other NR Cells
Default, SCENARIO_1~SCENARIO_11	Default, SCENARIO_1~SCENARIO_11
SCENARIO_12~SCENARIO_16	SCENARIO_12~SCENARIO_16

- When both LTE and NR cells are configured on a 32T32R RF module, these cells must meet the following requirements. Otherwise, cell coverage performance is affected, and if the **CARR_BEAM_CONFIG_DIFF_ALM_SW** option of the **gNBOamParam.AlarmSwitch** parameter is selected, the base station will report **ALM-29871 NR DU Cell TRP Capability Degraded** with the alarm cause being **Inconsistent beam configurations between co-sector carriers**.
 - The **NRDUCellTrpBeam.Tilt** parameter values of all NR cells must be the same as the **BfAnt.Tilt** parameter values of all LTE cells.
 - The coverage scenarios of NR and LTE cells must meet the requirements in [Table 5-5](#). For details about LTE cell coverage scenarios, see *Massive MIMO Basics (TDD)* in *eRAN Feature Documentation*.

Table 5-5 Requirements for the coverage scenarios of NR and LTE cells

NR Cell Coverage Scenario	LTE Cell Coverage Scenario
Default, SCENARIO_1~SCENARIO_11	SCENARIO_1~SCENARIO_9, SCENARIO_14~SCENARIO_17
SCENARIO_12~SCENARIO_16	SCENARIO_10~SCENARIO_13

5.1.1.2 MetaAAUs

Table 5-6 lists the MetaAAU models and supported 3D coverage scenarios.

Table 5-6 MetaAAU models and supported 3D coverage scenarios

RF Module	Band	TX/RX Mode	Supported 3D Coverage Scenarios
AAU5368md	2300 MHz + 3500 MHz	32T32R	Table 5-7
AAU5368md	2600 MHz + 3600 MHz	32T32R	Table 5-8
AAU5636m	3500 MHz	64T64R	Table 5-9
AAU5636m	3600 MHz	64T64R	Table 5-9
AAU5636m	3700 MHz	64T64R	Table 5-9
AAU5636m	2600 MHz	64T64R	Table 5-11
AAU5639m	3500 MHz	64T64R	Table 5-9
AAU5668m	3600 MHz	64T64R	Table 5-10
AAU5685md	2300 MHz + 2600 MHz	64T64R	Table 5-12

Table 5-7 3D coverage scenarios supported by the AAU5368md (2300 MHz + 3500 MHz)

Band	Scenario ID	Horizontal 3 dB Beamwidth	Vertical 3 dB Beamwidth	Tilt Adjustment Range	Azimuth Adjustment Range
2300 MHz	SCENARIO_1	110°	6°	0°~9°	0°
	SCENARIO_6	110°	12°	0°~9°	0°

Band	Scenario ID	Horizontal 3 dB Beamwidth	Vertical 3 dB Beamwidth	Tilt Adjustment Range	Azimuth Adjustment Range
	SCENARIO_7	90°	12°	0°~9°	-10°~10°
	SCENARIO_8	65°	12°	0°~9°	-22°~22°
	SCENARIO_12	110°	25°	6°	0°
	SCENARIO_13	65°	25°	6°	-22°~22°
	SCENARIO_14	45°	25°	6°	-32°~32°
	SCENARIO_15	25°	25°	6°	-42°~42°
	SCENARIO_36 ^a	105°	6°	0°~9°	0°
	CUSTOMIZED_BEAM_SCENARIO	15° or 30° for a single beam	6°	0°~9°	-47°~47°
3500 MHz	SCENARIO_8	65°	9°	0°~9°	-22°~22°
	SCENARIO_13	65°	17°	6°	-22°~22°
	SCENARIO_14	45°	17°	6°	-32°~32°
	SCENARIO_15	25°	17°	6°	-42°~42°
	SCENARIO_31	85°	4.5°	0°~9°	0°
a: This scenario has the same horizontal and vertical 3 dB beamwidths as the default coverage scenario but covers a different area. Perform configuration under the guidance of Huawei network optimization and planning engineers.					

Table 5-8 3D coverage scenarios supported by the AAU5368md (2600 MHz + 3600 MHz)

Band	Scenario ID	Horizontal 3 dB Beamwidth	Vertical 3 dB Beamwidth	Tilt Adjustment Range	Azimuth Adjustment Range
2600 MHz	SCENARIO_1	110°	6°	0°~9°	0°
	SCENARIO_6	110°	12°	0°~9°	0°
	SCENARIO_7	90°	12°	0°~9°	-10°~10°
	SCENARIO_8	65°	12°	0°~9°	-22°~22°
	SCENARIO_12	110°	25°	6°	0°
	SCENARIO_13	65°	25°	6°	-22°~22°
	SCENARIO_14	45°	25°	6°	-32°~32°
	SCENARIO_15	25°	25°	6°	-42°~42°
	CUSTOMIZED_BEAM_SCENARIO	15° or 30° for a single beam	6°	0°~9°	-47°~47°
3600 MHz	SCENARIO_8	65°	9°	0°~9°	-22°~22°
	SCENARIO_13	65°	17°	6°	-22°~22°
	SCENARIO_14	45°	17°	6°	-32°~32°
	SCENARIO_15	25°	17°	6°	-42°~42°
	SCENARIO_31	85°	4.5°	0°~9°	0°

Table 5-9 3D coverage scenarios supported by the AAU5636m (3500 MHz, 3600 MHz, or 3700 MHz)/AAU5639m (3500 MHz)

Scenario ID	Horizontal 3 dB Beamwidth	Vertical 3 dB Beamwidth	Tilt Adjustment Range	Azimuth Adjustment Range
SCENARIO_1	110°	6°	-2°~13°	0°
SCENARIO_2	90°	6°	-2°~13°	-10°~10°
SCENARIO_3	65°	6°	-2°~13°	-22°~22°
SCENARIO_4	45°	6°	-2°~13°	-32°~32°
SCENARIO_5	25°	6°	-2°~13°	-42°~42°
SCENARIO_6	110°	12°	3°, 6°, 9°	0°
SCENARIO_7	90°	12°	3°, 6°, 9°	-10°~10°
SCENARIO_8	65°	12°	3°, 6°, 9°	-22°~22°
SCENARIO_9	45°	12°	3°, 6°, 9°	-32°~32°
SCENARIO_10	25°	12°	3°, 6°, 9°	-42°~42°
SCENARIO_11	15°	12°	3°, 6°, 9°	-47°~47°
SCENARIO_12	110°	25°	6°	0°
SCENARIO_13	65°	25°	6°	-22°~22°
SCENARIO_14	45°	25°	6°	-32°~32°
SCENARIO_15	25°	25°	6°	-42°~42°
SCENARIO_16	15°	25°	6°	-47°~47°
CUSTOMIZED_BEAM_SCENARIO	15° or 30° for a single beam	6°	2°~8°	-47°~47°

Table 5-10 3D coverage scenarios supported by the AAU5668m (3600 MHz)

Scenario ID	Horizontal 3 dB Beamwidth	Vertical 3 dB Beamwidth	Tilt Adjustment Range	Azimuth Adjustment Range
SCENARIO_1	110°	6°	-2°~13°	0°
SCENARIO_2	90°	6°	-2°~13°	-10°~10°
SCENARIO_3	65°	6°	-2°~13°	-22°~22°
SCENARIO_4	45°	6°	-2°~13°	-32°~32°
SCENARIO_5	25°	6°	-2°~13°	-42°~42°

Scenario ID	Horizontal 3 dB Beamwidth	Vertical 3 dB Beamwidth	Tilt Adjustment Range	Azimuth Adjustment Range
SCENARIO_6	110°	12°	3°, 6°, 9°	0°
SCENARIO_7	90°	12°	3°, 6°, 9°	-10°~10°
SCENARIO_8	65°	12°	3°, 6°, 9°	-22°~22°
SCENARIO_9	45°	12°	3°, 6°, 9°	-32°~32°
SCENARIO_10	25°	12°	3°, 6°, 9°	-42°~42°
SCENARIO_11	15°	12°	3°, 6°, 9°	-47°~47°
SCENARIO_12	110°	25°	6°	0°
SCENARIO_13	65°	25°	6°	-22°~22°
SCENARIO_14	45°	25°	6°	-32°~32°
SCENARIO_15	25°	25°	6°	-42°~42°
SCENARIO_16	15°	25°	6°	-47°~47°
SCENARIO_37 ^a	105°	6°	-2°~13°	0°
CUSTOMIZED_BEAM_SCENARIO	15° or 30° for a single beam	6°	2°~8°	-47°~47°
a: This scenario has the same horizontal and vertical 3 dB beamwidths as the Default coverage scenario but has a different upper sidelobe size. Perform configuration under the guidance of Huawei network optimization and planning engineers.				

Table 5-11 3D coverage scenarios supported by the AAU5636m (2600 MHz)

Scenario ID	Horizontal 3 dB Beamwidth	Vertical 3 dB Beamwidth	Tilt Adjustment Range	Azimuth Adjustment Range
SCENARIO_1	110°	4.5°	-2°~13°	0°
SCENARIO_2	90°	4.5°	-2°~13°	-10°~10°
SCENARIO_3	65°	4.5°	-2°~13°	-22°~22°
SCENARIO_4	45°	4.5°	-2°~13°	-32°~32°
SCENARIO_5	25°	4.5°	-2°~13°	-42°~42°
SCENARIO_6	110°	9°	0°~9°	0°
SCENARIO_7	90°	9°	0°~9°	-10°~10°

Scenario ID	Horizontal 3 dB Beamwidth	Vertical 3 dB Beamwidth	Tilt Adjustment Range	Azimuth Adjustment Range
SCENARIO_8	65°	9°	0°~9°	-22°~22°
SCENARIO_9	45°	9°	0°~9°	-32°~32°
SCENARIO_10	25°	9°	0°~9°	-42°~42°
SCENARIO_11	15°	9°	0°~9°	-47°~47°
SCENARIO_12	110°	25°	6°	0°
SCENARIO_13	65°	25°	6°	-22°~22°
SCENARIO_14	45°	25°	6°	-32°~32°
SCENARIO_15	25°	25°	6°	-42°~42°
SCENARIO_16	15°	25°	6°	-47°~47°
SCENARIO_31	105°	6°	-2°~13°	0°
CUSTOMIZED_BEAM_SCENARIO	15° or 30° for a single beam	4.5°	3°~8°	-47°~47°

Table 5-12 3D coverage scenarios supported by the AAU5685md (2300 MHz + 2600 MHz)

Scenario ID	Horizontal 3 dB Beamwidth	Vertical 3 dB Beamwidth	Tilt Adjustment Range	Azimuth Adjustment Range
SCENARIO_1	110°	6°	-2°~13°	0°
SCENARIO_2	90°	6°	-2°~13°	-10°~10°
SCENARIO_3	65°	6°	-2°~13°	-22°~22°
SCENARIO_4	45°	6°	-2°~13°	-32°~32°
SCENARIO_5	25°	6°	-2°~13°	-42°~42°
SCENARIO_6	110°	12°	0°~9°	0°
SCENARIO_7	90°	12°	0°~9°	-10°~10°
SCENARIO_8	65°	12°	0°~9°	-22°~22°
SCENARIO_9	45°	12°	0°~9°	-32°~32°
SCENARIO_10	25°	12°	0°~9°	-42°~42°
SCENARIO_11	15°	12°	0°~9°	-47°~47°

Scenario ID	Horizontal 3 dB Beamwidth	Vertical 3 dB Beamwidth	Tilt Adjustment Range	Azimuth Adjustment Range
SCENARIO_12	110°	25°	6°	0°
SCENARIO_13	65°	25°	6°	-22°~22°
SCENARIO_14	45°	25°	6°	-32°~32°
SCENARIO_15	25°	25°	6°	-42°~42°
SCENARIO_16	15°	25°	6°	-47°~47°
CUSTOMIZED_BEAM_SCENARIO	15° or 30° for a single beam	6°	3°~8°	-47°~47°

The coverage scenario configuration requirements for MetaAAUs are as follows:

- Different AAUs support different coverage scenarios, which can be queried by running the **DSP NRDUCELLTRP** command. The base station reports ALM-29871 NR DU Cell TRP Capability Degraded, with the **Specific Problem** being **Beam configuration error** (that is, the coverage scenario, tilt angle, or azimuth exceeds the capability of the RF module) in the following situations:
 - An unsupported coverage scenario is configured. If it is configured for a cell before the cell is set up, the default coverage scenario will take effect for the cell. If the coverage scenario of a cell is changed to an unsupported one when the cell is in the active state, the change will not take effect and the last valid value will be retained. If the coverage scenario configured for a cell is not supported by a new AAU that replaces an old one, the default coverage scenario will take effect for the cell.
 - An unsupported tilt or azimuth is configured. If it is configured for a cell before the cell is set up, the valid tilt or azimuth closest to the configured value will take effect for the cell. (If multiple values are closest to the configured one, the minimum value will take effect.) If the tilt or azimuth of a cell is changed to an unsupported one when the cell is in the active state, the change will not take effect and the last valid value will be retained. If the tilt or azimuth configured for a cell is not supported by a new AAU that replaces an old one, the valid tilt or azimuth closest to the configured value will take effect for the cell. (If multiple values are closest to the configured one, the minimum value will take effect.)
- When multiple NR cells are configured on a MetaAAU, these cells must meet the following requirements. Otherwise, cell coverage performance is affected, and if the **CARR_BEAM_CONFIG_DIFF_ALM_SW** option of the **gNBOamParam.AlarmSwitch** parameter is selected, the base station will report **ALM-29871 NR DU Cell TRP Capability Degraded** with the alarm cause being **Inconsistent beam configurations between co-sector carriers**.
 - The **NRDUCellTrpBeam.Tilt** parameter values of all cells must be the same.

- The coverage scenarios of multiple NR cells must meet the requirements in [Table 5-13](#).

Table 5-13 Requirements for the coverage scenarios of multiple NR cells

Coverage Scenario of One NR Cell	Coverage Scenarios of Other NR Cells
Default, SCENARIO_1~SCENARIO_11	Default, SCENARIO_1~SCENARIO_11
SCENARIO_12~SCENARIO_16	SCENARIO_12~SCENARIO_16

- When both LTE and NR cells are configured on a MetaAAU, these cells must meet the following requirements. Otherwise, cell coverage performance is affected, and if the **CARR_BEAM_CONFIG_DIFF_ALM_SW** option of the **gNBOamParam.AlarmSwitch** parameter is selected, the base station will report **ALM-29871 NR DU Cell TRP Capability Degraded** with the alarm cause being **Inconsistent beam configurations between co-sector carriers**.
 - The **NRDUCellTrpBeam.Tilt** parameter values of all NR cells must be the same as the **BfAnt.Tilt** parameter values of all LTE cells.
 - The coverage scenarios of NR and LTE cells must meet the requirements in [Table 5-14](#). For details about LTE cell coverage scenarios, see *Massive MIMO Basics (TDD)* in *eRAN Feature Documentation*.

Table 5-14 Requirements for the coverage scenarios of NR and LTE cells

NR Cell Coverage Scenario	LTE Cell Coverage Scenario
Default, SCENARIO_1~SCENARIO_11	SCENARIO_1~SCENARIO_9, SCENARIO_14~SCENARIO_18
SCENARIO_12~SCENARIO_16	SCENARIO_10~SCENARIO_13

5.1.2 TDD 8T Cells

The 3D coverage scenario of a cell is specified by the **NRDUCellTrpBeam.CoverageScenario** parameter.

For the 3D coverage scenarios supported in TDD 8T cells served by Book RRUs or common RRUs with external antennas, see [Cells Served by Book RRUs or Common RRUs with External Antennas](#). For the 3D coverage scenarios supported in TDD 8T cells served by Easy Macro AAUs with internal antennas, see [Cells Served by Easy Macro AAUs with Internal Antennas](#).

Cells Served by Book RRUs or Common RRUs with External Antennas

Coverage scenarios SCENARIO_3 to SCENARIO_5 are supported. If the 8T8R ATD4516R8 antenna is used, the tilt and azimuth adjustment ranges listed in [Table 5-15](#) apply. If other types of external antennas are used, the tilt and azimuth adjustment ranges are determined by the external antennas.

Table 5-15 3D coverage scenarios of broadcast beams for TDD 8T cells

Scenario ID	Horizontal 3 dB Beamwidth	Vertical 3 dB Beamwidth	Tilt Adjustment Range	Azimuth Adjustment Range
SCENARIO_3	65°	6°	0°	-10°~10°
SCENARIO_4	45°	6°	0°	-22°~22°
SCENARIO_5	25°	6°	0°	-32°~32°
If any of coverage scenarios SCENARIO_3 to SCENARIO_5 is used for an 8T8R cell operating in band n38 or n40, the SCENARIO_BEAM_OPT_SW option of the NRDUCellBeamAlgo.WeightAlgoSwitch parameter must be selected for the cell.				

Cells Served by Easy Macro AAUs with Internal Antennas

Coverage scenarios SCENARIO_29, SCENARIO_30, and SCENARIO_32 are supported. The AAU5246 must be used. The details of each coverage scenario depend on the value of the **RVDSUBUNIT.VBEAMWIDTH** parameter, as described in [Table 5-16](#).

Table 5-16 3D coverage scenarios supported by the AAU5246 (2600 MHz)

Scenario ID	Value of RVDSUBUNIT.VBEAMWIDTH	Horizontal 3 dB Beamwidth	Vertical 3 dB Beamwidth	Tilt Adjustment Range ^a	Azimuth Adjustment Range ^b
SCENARIO_29	200	35°	21°	2°~12°	0°
	400	35°	43°	7°	0°
SCENARIO_30 ^c	200	65°	10°	2°~12°	0°
Note: a: The tilt is specified by the RETSUBUNIT.TILT parameter. The NRDUCellTrpBeam.Tilt parameter must be set to 0. b: The azimuth cannot be adjusted. The NRDUCellTrpBeam.Azimuth parameter must be set to 0. c: When RVDSUBUNIT.VBEAMWIDTH is set to a value other than 200 in SCENARIO_30, the default coverage scenario (detailed in Cells Served by Easy Macro AAUs with Internal Antennas in 4.1.1.2 TDD 2T/4T/8T Cells) takes effect.					

5.1.3 Other Coverage Improvement Functions

The following functions can be used to improve cell coverage:

- Broadcast beam densification: This function is controlled by the **SCENARIO_BEAM_DENSIFY_ALGO_SW** option of the **NRDUCellTrpBeam.ScenarioBeamAlgoSw** parameter. If this function is enabled, broadcast beams are sent based on the maximum number of beams supported under the current slot configuration. This improves cell coverage. This function is supported only when the **NRDUCellTrpBeam.CoverageScenario** parameter is set to **DEFAULT** or a value in the range of **SCENARIO_1** to **SCENARIO_16**.
- Power aggregation: This function improves cell edge coverage by using the maximum SSB transmit power offset specified by the **NRDUCellTrpBeam.MaxSsbPwrOffset** parameter. For details, see *Power Control*.

5.2 Network Analysis

5.2.1 Benefits

Different broadcast beam scenarios can be selected based on network conditions and UE distribution. The tilt and azimuth adjustments can reduce overshoot coverage, reduce inter-cell interference, and improve coverage performance in multiple scenarios, such as public squares or high-rise buildings.

5.2.2 Impacts

The impacts between this function and other functions are described in [5.3.2.3 Function Impacts](#).

After the broadcast beam coverage scenario of a cell is adjusted, there will be a change in the cell coverage scope and accordingly changes in indicators such as the UE quantity, UE throughput, access, handover, and service drops in the cell.

5.3 Requirements

5.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-010100	3D Coverage Pattern	NR0SBSC3DC00	per Cell

Feature ID	Feature Name	Model	Sales Unit
Note: Coverage scenario SCENARIO_31/SCENARIO_37 does not require the license for 3D Coverage Pattern. For the license control item NR0SBSC3DC00: • If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency. • If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation, the cell cannot be activated again after the feature is enabled and the parameters that cause automatic cell reset are modified.			

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

None

5.3.2.2 Mutually Exclusive Functions

Function Name	Function Switch	Reference	Description
Distributed massive MIMO	DM_MIMO_SERVICE_SWITCH option of the NRDUCellAlgoSwitch.DmMimoSwitch parameter	<i>Distributed Antenna Solutions (Low-Frequency TDD)</i>	The NRDUCellTrpBeam.CoverageScenario parameter must be set to DEFAULT for distributed massive MIMO cells.

5.3.2.3 Function Impacts

Function Name	Function Switch	Reference	Description
Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	It is recommended that the settings of the SCENARIO_BEAM_DENSIFY_ALGO_SW option of the NRDUCellTrpBeam.ScenarioBeamAlgoSw parameter be the same for all TRPs of a hyper cell to prevent UE access exceptions and improve system reliability.
Cell Combination	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE	<i>Cell Combination</i>	It is recommended that the settings of the SCENARIO_BEAM_DENSIFY_ALGO_SW option of the NRDUCellTrpBeam.ScenarioBeamAlgoSw parameter be the same for all TRPs of a combined cell to prevent UE access exceptions and improve system reliability.
Fusion Cell	NRDUCell.NrDuCellNetworkingMode set to FUSION_CELL	<i>Fusion Cell (Low-Frequency TDD)</i>	It is recommended that the settings of the following parameters be consistent between the two TRPs serving a cell enabled with Fusion Cell to obtain complete coverage gains: • NRDUCellTrpBeam.CoverageScenario • SCENARIO_BEAM_DENSIFY_ALGO_SW option of the NRDUCellTrpBeam.ScenarioBeamAlgoSw parameter • NRDUCellTrpBeam.Azimuth and NRDUCellTrpBeam.Tilt

Function Name	Function Switch	Reference	Description
SSB UE Compatibility Switch	SSB_UE_COM P_ALGO_SW option of the NRDUCellAlgoSwitch.AlgoCompatibilitySwitchExt parameter	None	When the SSB UE Compatibility Switch is turned on, the number of transmitted SSB beams decreases. As a result, the SSB coverage shrinks. If this switch is turned on in customized broadcast beam coverage scenarios, the SSB coverage may shrink more severely.
NR low power consumption mode	LOW_PWR_MODE_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Green Site</i>	Changes in beams will cause termination of low power consumption mode when both low power consumption mode and broadcast beam densification are enabled and a customized backup power saving policy is used in low power consumption mode.

5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

This function requires NR TDD-capable low-frequency RF modules with at least 8T8R.

5.3.4 Cells

None

5.4 Operation and Maintenance

5.4.1 Data Configuration

5.4.1.1 Data Preparation

Table 5-17 describes the parameters used for function activation.

Table 5-17 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Coverage Scenario	NRDUCellTrpBeam.Coverage Scenario	Based on the information such as engineering parameters on the live network and AAU antenna files, perform professional network planning and optimization to analyze the coverage and determine the most suitable coverage scenario, tilt, and azimuth. The AAU antenna files provide antenna gains in different positions and different coverage scenarios. If the coverage scenario needs to be customized, configure information about each beam using the NRDUCellTrpCust-Beam MO and then set this parameter to CUSTOMIZED_BEAM_SCENARIO.

Parameter Name	Parameter ID	Setting Notes
Tilt	NRDUCellTrpBeam.Tilt	Set this parameter based on the network plan. If this parameter is set to the default value 255 in scenarios with 32T32R or higher, the tilt is 6°. The antenna tilt is the value specified by this parameter plus the mechanical tilt. For RF modules that support the RET system, tilt adjustment is implemented by phase shifters. Changing the value of this parameter will affect the tilts of all channels and signals. All 32T32R modules support the RET system. For details, see section "AAU RET System" in the related hardware description. No 64T64R modules support the RET system.
Azimuth	NRDUCellTrpBeam.Azimuth	Set this parameter based on the network plan.
Scenario-based Beam Algorithm Switch	NRDUCellTrpBeam.ScenarioBeamAlgoSw	If coverage scenarios are different among neighboring cells, the cells will interfere with each other and downlink throughput will decrease. It is recommended that the SCENARIO_BEAM_DENSIFY_ALGO_SW option be selected to reduce interference.
Max SSB Power Offset	NRDUCellTrpBeam.MaxSsbPowerOffset	It is recommended that a required offset be configured if the PBCH or SS power is limited, for example, in scenarios with a large inter-site distance or using supplementary uplink (SUL).
Beam Id	NRDUCellTrpCust-Beam.BeamId	Set this parameter under the guidance of Huawei engineers based on the actual coverage scenario.

Parameter Name	Parameter ID	Setting Notes
Tilt	NRDUCellTrpCustBeam.Tilt	Set this parameter under the guidance of Huawei engineers based on the actual coverage scenario.
Azimuth	NRDUCellTrpCust-Beam.Azimuth	Set this parameter under the guidance of Huawei engineers based on the actual coverage scenario.
SSB Beam Horizontal Width	NRDUCellTrpCust-Beam.SsbBeamHorizWidth	Set this parameter under the guidance of Huawei engineers based on the actual coverage scenario.
Max SSB Power Offset	NRDUCellTrpCust-Beam.MaxSsbPwrOffset	Set this parameter under the guidance of Huawei engineers based on the actual coverage scenario.
Vertical BeamWidth	RVDSUBUNIT.VBEAMWIDTH	Set this parameter based on the network plan.
Tilt	RETSUBUNIT.TILT	Set this parameter based on the network plan.
Weight Algorithm Switch	NRDUCellBeamAlgo.WeightAlgoSwitch	For an 8T8R cell operating in band n38 or n40 or a 64T64R cell served by the AAU5614 (2600 MHz), select the SCENARIO_BEAM_OPT_SW option of this parameter if any of coverage scenarios SCENARIO_3 to SCENARIO_5 is used for the cell.

NOTICE

It is recommended that the **NRDUCellTrpBeam.CoverageScenario** parameter or the **SCENARIO_BEAM_DENSIFY_ALGO_SW** option of the **NRDUCellTrpBeam.ScenarioBeamAlgoSw** parameter be adjusted during off-peak hours. This is because the service performance of UEs in connected mode or the access performance will be affected within 5 minutes after the adjustment. The related indicators include the UE throughput, access KPIs, handover KPIs, and service drop KPIs.

5.4.1.2 Using MML Commands

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

Run the **DSP NRDUCELLTRP** command to query the coverage scenario, tilt adjustment range, and azimuth adjustment range supported by the RF module in use before configuration.

TDD 64T/32T cells

- For coverage scenarios SCENARIO_1 to SCENARIO_16:

```
//Setting the broadcast beam coverage scenario, tilt, and azimuth
MOD NRDUCELLTRPBEAM: NrDuCellTrpld=1, CoverageScenario=SCENARIO_1, Tilt=255, Azimuth=0;
//Enabling broadcast beam densification and power aggregation
MOD NRDUCELLTRPBEAM: NrDuCellTrpld=1, CoverageScenario=SCENARIO_1,
ScenarioBeamAlgoSw=SCENARIO_BEAM_DENSIFY_ALGO_SW-1, MaxSsbPwrOffset=3;
```
- For coverage scenario CUSTOMIZED_BEAM_SCENARIO:

```
//Configuring customized broadcast beams
ADD NRDUCELLTRPCUSTBEAM: NrDuCellTrpld=1, BeamId=BEAM_0, Tilt=6, Azimuth=-40,
SsbBeamHorizWidth=WIDTH_30, MaxSsbPwrOffset=3;
ADD NRDUCELLTRPCUSTBEAM: NrDuCellTrpld=1, BeamId=BEAM_1, Tilt=6, Azimuth=-22;
ADD NRDUCELLTRPCUSTBEAM: NrDuCellTrpld=1, BeamId=BEAM_2, Tilt=6, Azimuth=-7;
ADD NRDUCELLTRPCUSTBEAM: NrDuCellTrpld=1, BeamId=BEAM_3, Tilt=6, Azimuth=7;
MOD NRDUCELLTRPBEAM: NrDuCellTrpld=1, CoverageScenario=CUSTOMIZED_BEAM_SCENARIO,
Tilt=255, Azimuth=0, MaxSsbPwrOffset=0;
```

TDD 8T cells

- For coverage scenarios SCENARIO_3 to SCENARIO_5:

```
//Enabling scenario-based beam optimization for an 8T8R cell operating in band n38 or n40
MOD NRDUCELLBEAMALGO: NrDuCellId=1, WeightAlgoSwitch=SCENARIO_BEAM_OPT_SW-1;
//Setting the broadcast beam coverage scenario, tilt, and azimuth
MOD NRDUCELLTRPBEAM: NrDuCellTrpld=1, CoverageScenario=SCENARIO_3, Tilt=255, Azimuth=0;
//Enabling broadcast beam densification and power aggregation
MOD NRDUCELLTRPBEAM: NrDuCellTrpld=1, CoverageScenario=SCENARIO_3,
ScenarioBeamAlgoSw=SCENARIO_BEAM_DENSIFY_ALGO_SW-1, MaxSsbPwrOffset=3;
```
- For coverage scenarios SCENARIO_29 and SCENARIO_30 (where the RF module must be AAU5246):

```
//Setting the broadcast beam coverage scenario, tilt, and azimuth
MOD RVDSUBUNIT: DEVICENO=3, SUBUNITNO=1, VBEAMWIDTH=200;
MOD NRDUCELLTRPBEAM: NrDuCellTrpld=1, CoverageScenario=SCENARIO_29;
MOD RETSUBUNIT: DEVICENO=0, SUBUNITNO=1, TILT=80;
//Enabling broadcast beam densification and power aggregation
MOD NRDUCELLTRPBEAM: NrDuCellTrpld=1, CoverageScenario=SCENARIO_29,
ScenarioBeamAlgoSw=SCENARIO_BEAM_DENSIFY_ALGO_SW-1, MaxSsbPwrOffset=3;
```

Deactivation Command Examples

TDD 64T/32T cells

- For coverage scenarios SCENARIO_1 to SCENARIO_16:
 //Restoring the default coverage scenario
 MOD NRDUCELLTRPBEAM: NrDuCellTrpld=1, CoverageScenario=DEFAULT;
 //Disabling broadcast beam densification and power aggregation
 MOD NRDUCELLTRPBEAM: NrDuCellTrpld=1, CoverageScenario=DEFAULT,
 ScenarioBeamAlgoSw=SCENARIO_BEAM_DENSIFY_ALGO_SW-0, MaxSsbPwrOffset=0;
- For coverage scenario CUSTOMIZED_BEAM_SCENARIO:
 //Removing the customized broadcast beam coverage scenario and restoring the default coverage scenario
 RMV NRDUCELLTRPCUSTBEAM: NrDuCellTrpld=1, BeamId=BEAM_0;
 RMV NRDUCELLTRPCUSTBEAM: NrDuCellTrpld=1, BeamId=BEAM_1;
 RMV NRDUCELLTRPCUSTBEAM: NrDuCellTrpld=1, BeamId=BEAM_2;
 RMV NRDUCELLTRPCUSTBEAM: NrDuCellTrpld=1, BeamId=BEAM_3;
 MOD NRDUCELLTRPBEAM: NrDuCellTrpld=1, CoverageScenario=DEFAULT;

TDD 8T cells

- For coverage scenarios SCENARIO_3 to SCENARIO_5:
 //Restoring the default coverage scenario
 MOD NRDUCELLTRPBEAM: NrDuCellTrpld=1, CoverageScenario=DEFAULT;
 //Disabling broadcast beam densification and power aggregation
 MOD NRDUCELLTRPBEAM: NrDuCellTrpld=1, CoverageScenario=DEFAULT,
 ScenarioBeamAlgoSw=SCENARIO_BEAM_DENSIFY_ALGO_SW-0, MaxSsbPwrOffset=0;
 //Disabling scenario-based beam optimization
 MOD NRDUCELLBEAMALGO: NrDuCellId=1, WeightAlgoSwitch=SCENARIO_BEAM_OPT_SW-0;
- For coverage scenarios SCENARIO_29 and SCENARIO_30 (where the RF module must be AAU5246):
 //Restoring the default coverage scenario
 MOD NRDUCELLTRPBEAM: NrDuCellTrpld=1, CoverageScenario=DEFAULT;
 //Disabling broadcast beam densification and power aggregation
 MOD NRDUCELLTRPBEAM: NrDuCellTrpld=1, CoverageScenario=DEFAULT,
 ScenarioBeamAlgoSw=SCENARIO_BEAM_DENSIFY_ALGO_SW-0, MaxSsbPwrOffset=0;

5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.2 Activation Verification

Run the **DSP NRDUCELLTRP** command to check the value of **Actual Beam Scenario**. If this value is consistent with the configured value, the function has taken effect.

For 32T/64T cells, you can run an MML command to query information about each SSB beam. For non-customized broadcast beam coverage scenarios, the command is **LST NRDUCELLTRPSUBBEAM**. For customized broadcast beam coverage scenarios, the command is **LST NRDUCELLTRPCUSTBEAM**.

For 32T/64T cells, you can use the counters listed in [Table 5-18](#) to observe the number of UEs and traffic volume under the optimal SSB beam.

Table 5-18 Number of UEs and traffic volume under the optimal SSB beam

Counter ID	Counter Name	Description
1911827216	N.User.OptimalSSBBeam.Avg	Average number of UEs under the optimal SSB beam
1911827224	N.MAC.ThpVol.UL.OptimalSSB Beam	Total uplink traffic volume at the MAC layer under the optimal SSB beam
1911827225	N.MAC.ThpVol.DL.OptimalSSB Beam	Total downlink traffic volume at the MAC layer under the optimal SSB beam

 NOTE

The optimal SSB beam of UEs is determined based on SRS measurement and may not be the SSB beam actually used by the UEs for access. The measured number of UEs and traffic volume under some SSB beams may be 0, which does not indicate the actual coverage of these SSB beams.

5.4.3 Network Monitoring

Use **N.User.RRCCConn.Avg** to observe the coverage changes in different scenarios.

6 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

7 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

8 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

9 Reference Documents

- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- 3GPP TS 38.213: "NR; Physical layer procedures for control"
- 3GPP TS 38.331: "NR; Radio Resource Control (RRC) protocol specification"
- 3GPP TS 38.211: "NR; Physical channels and modulation"
- *mmWave Beam Management (High-Frequency TDD)*
- *MIMO (TDD)*
- *Carrier Aggregation*
- *Standards Compliance*
- *Distributed Antenna Solutions (Low-Frequency TDD)*
- *Channel Management*
- *Hyper Cell*
- *Cell Combination*
- *Energy Conservation and Emission Reduction*
- *Power Control*
- *Fusion Cell (Low-Frequency TDD)*
- *Green Site*
- *License Control Item Lists*
- *Emergency Communications Solution (Low-Frequency TDD)*
- *Technical Specifications in 3900 & 5900 Series Base Station Product Documentation*